

$X = \text{input data}$

$y = \text{output data}$

We need $p\left(\frac{y}{x}\right)$

$$P(X \cap Y) = P\left(\frac{X}{Y}\right) * P(Y) \text{ or } P\left(\frac{Y}{X}\right) * P(X)$$

$$P\left(\frac{X}{Y}\right) * P(Y) = P\left(\frac{Y}{X}\right) * P(X) \quad \text{====} A * B = C * D : C = A * \frac{B}{D}$$

$$P\left(\frac{Y}{X}\right) = \frac{P\left(\frac{X}{Y}\right) * P(Y)}{P(X)}$$

Process Of Model Development

$P(X) = \text{Probability of } X = \text{Probability of Getting Input data}$

It is available from Data set

$P(Y) = \text{Probability of } Y = \text{Probability of Getting Output data}$

It is available from Data set

while we are doing train the model , both Input and Output available

$$P\left(\frac{X}{Y}\right) = \text{Probability of } X \text{ when } Y \text{ is there} =$$

Probability of Getting Input data when Output data available

It is available from Data set

After Model Development

$$P\left(\frac{Y}{X}\right) = \text{Probability of } Y \text{ when } X \text{ is there}$$

Sir last line what is y and what is x

Means train data and prediction data?

We are passing input ==== we are expecting output from model

$P\left(\frac{Y}{X}\right)$ = Probability of Y when X is there

what is exactly meaning of Probability of Y

in a classification problem we have multiple labels are there

Buy a computer problem , we have two classes or two labels

- **yes**
- **No**

we already know that ML model will give first probability

which probability has highest one that is the answer

$P\left(\frac{Y = \text{yes}}{X}\right)$ = Probability of Y = yes when X is there

$P\left(\frac{Y = \text{no}}{X}\right)$ = Probability of Y = no when X is there

$\max\{P\left(\frac{Y=\text{yes}}{X}\right), P\left(\frac{Y=\text{No}}{X}\right)\}$ ===== Final output

is there may be more than one input features? yes

$$P(X \cap Y) = P\left(\frac{X}{Y}\right) * P(Y) \text{ or } P\left(\frac{Y}{X}\right) * P(X)$$

$$P\left(\frac{X}{Y}\right) * P(Y) = P\left(\frac{Y}{X}\right) * P(X)$$

$$P\left(\frac{Y}{X}\right) = \frac{P\left(\frac{X}{Y}\right) * P(Y)}{P(X)}$$

$$P\left(\frac{Y = \text{yes}}{X}\right) = \frac{P\left(\frac{X}{Y = \text{yes}}\right) * P(Y = \text{yes})}{P(X)}$$

$$P\left(\frac{Y = \text{No}}{X}\right) = \frac{P\left(\frac{X}{Y = \text{No}}\right) * P(Y = \text{No})}{P(X)}$$

$$P\left(\frac{Y = \text{yes}}{X}\right) = P\left(\frac{X}{Y = \text{yes}}\right) * P(Y = \text{yes}) ===== (A)$$

$$P\left(\frac{Y = \text{No}}{X}\right) = P\left(\frac{X}{Y = \text{No}}\right) * P(Y = \text{No}) ===== (B)$$

$$P\left(\frac{Y = \text{yes}}{X}\right) = 0.7, \quad P\left(\frac{Y = \text{no}}{X}\right) = 0.3$$

We need to find two values

$$P\left(\frac{Y = \text{yes}}{X}\right) \text{ and } P\left(\frac{Y = \text{No}}{X}\right)$$

$$P\left(\frac{Y = \text{yes}}{X}\right) = P\left(\frac{X}{Y = \text{yes}}\right) * P(Y = \text{yes}) ===== (A)$$

$$P\left(\frac{Y = \text{No}}{X}\right) = P\left(\frac{X}{Y = \text{No}}\right) * P(Y = \text{No}) ===== (B)$$

$X = (\text{age} = \text{youth}, \text{income} = \text{medium}, \text{student} = \text{yes}, \text{credit_rating} = \text{fair})$

Step – 1:

$$P\left(\frac{Y = \text{yes}}{X}\right) = P\left(\frac{X}{Y = \text{yes}}\right) * P(Y = \text{yes})$$

$$P\left(\frac{X = \{\text{age: youth, income: med, student: yes, cr: fair}\}}{Y = \text{yes}}\right)$$

Assumption in Naive Bayes:

Class conditional independence: all the columns are independent each other

we already seen a similar assumption in linear regression: No Multicollinearity

It is always an Important , columns should be independent each other

we already know that Independent is a spl case in multiplication theorem

$$P(A \cap B) = P(A) * P(B)$$

$$P\left(\frac{X = \{\text{age: youth, income: med, student: yes, cr: fair}\}}{Y = \text{yes}}\right) =$$

$$P\left(\frac{X: \text{age} = \text{youth}}{Y = \text{yes}}\right) * P\left(\frac{X: \text{income} = \text{med}}{Y = \text{yes}}\right) * P\left(\frac{X: \text{student} = \text{yes}}{Y = \text{yes}}\right) * P\left(\frac{X: \text{cr} = \text{fair}}{Y = \text{yes}}\right)$$

$$P\left(\frac{X: \text{age} = \text{youth}}{Y = \text{yes}}\right) = \text{out of total yes cases, how many are belongs to youth}$$

$$P\left(\frac{X: \text{age} = \text{youth}}{Y = \text{yes}}\right) = \frac{2}{9}$$

$$P\left(\frac{X: \text{income} = \text{med}}{Y = \text{yes}}\right) = \frac{4}{9}$$

$$P\left(\frac{X: student = yes}{Y = yes}\right) = \frac{6}{9}$$

$$P\left(\frac{X: cr = fair}{Y = yes}\right) = \frac{6}{9}$$

Before applying ML we will convert into Independent also PCA

$$P\left(\frac{X = \{age: youth, income: med, student: yes, cr: fair\}}{Y = yes}\right) =$$

$$\frac{2}{9} * \frac{4}{9} * \frac{6}{9} * \frac{6}{9} = \frac{288}{6561}$$

Step – 2:

$$P(Y = yes) = \frac{9}{14}$$

step – 3:

$$P\left(\frac{Y = yes}{X}\right) = P\left(\frac{X}{Y = yes}\right) * P(Y = yes)$$

$$P\left(\frac{Y = yes}{X}\right) = \frac{288}{6561} * \frac{9}{14} = 0.028$$

step – 4: No case:

$$P\left(\frac{X = \{age: youth, income: med, student: yes, cr: fair\}}{Y = no}\right) =$$

$$P\left(\frac{X: age = youth}{Y = no}\right) * P\left(\frac{X: income = med}{Y = no}\right) * P\left(\frac{X: student = yes}{Y = no}\right) * P\left(\frac{X: cr = fair}{Y = no}\right)$$

$$P\left(\frac{X: age = youth}{Y = no}\right) = \text{out of total no cases, how many are belongs to youth}$$

$$P\left(\frac{X: age = youth}{Y = no}\right) = \frac{3}{5}$$

$$P\left(\frac{X: income = med}{Y = no}\right) = \frac{2}{5}$$

$$P\left(\frac{X: student = yes}{Y = no}\right) = \frac{1}{5}$$

$$P\left(\frac{X: cr = fair}{Y = no}\right) = \frac{2}{5}$$

$$P\left(\frac{X = \{age: youth, income: med, student: yes, cr: fair\}}{Y = no}\right) =$$

$$\frac{3}{5} * \frac{2}{5} * \frac{1}{5} * \frac{2}{5} = \frac{12}{625}$$

step – 5: No case:

$$P(Y = No) = \frac{5}{14}$$

step – 6:

$$P\left(\frac{Y = no}{X}\right) = P\left(\frac{X}{Y = no}\right) * P(Y = no)$$

$$\frac{12}{625} * \frac{5}{14} = 0.007$$

Step – 7: Conclusion

$$P\left(\frac{Y = yes}{X}\right) = \frac{288}{6561} * \frac{9}{14} = 0.028$$

$$P\left(\frac{Y = no}{X}\right) = \frac{12}{625} * \frac{5}{14} = 0.007$$

$X = (age = youth, income = medium, student = yes, credit_rating = fair)$

$$P\left(\frac{Y = yes}{X}\right) > P\left(\frac{Y = no}{X}\right) : \text{The person buys a computer}$$

Note:

Two are different use cases , Both conditions are different

we are getting un normalized values

we are skipping the $p(x)$ denominator

$$P\left(\frac{Y = \text{yes}}{X}\right) + P\left(\frac{Y = \text{No}}{X}\right) = 1$$

draw back:

$$P\left(\frac{X = \{\text{age: youth, income: med, student: yes, cr: fair}\}}{Y = \text{yes}}\right) =$$

$$\frac{2}{9} * \frac{4}{9} * \frac{6}{9} * \frac{6}{9} = \frac{288}{6561}$$

Imagine that , **There is No observation for Yes cases when age = Youth**

age	Buys computer
Youth	No
Youth	No
Youth	No

$$\frac{0}{9} * \frac{4}{9} * \frac{6}{9} * \frac{6}{9} = \frac{0}{6561} = 0$$

We can say that final answer is NO (Not buy),

with out learning any patterns we are able to provide the prediction

Draw back is :

- In Naive bayes we are using probability ,which means we are taking a count of an experiment
- There might be a chance ,the count becomes zero
- Due to the assumption, the multiplication factor is zero
- So the entire calculation becomes zero

- it is not giving a chance how other feature are behaving
- There is no pattern learning

Laplacian correction

- some training database, D , containing 1000 tuples (Rows or observations),
 - we have 0 tuples(rows) with $income = low$,
 - 990 tuples with $income = medium$, and
 - 10 tuples with $income = high$.

A data set has 1000 observations, in that one column name : Income

Income has three labels

- low: 0
- Medium : 990
- High: 10

$$p(low) = \frac{0}{1000} = 0; \quad p(Medium) = \frac{990}{1000}, \quad p(High) = \frac{10}{1000}$$

Correction:

Add extra rows = number of labels

there three labels are there i.e. low, high, medium

Extra three rows we need to add

Age	Student	Income	cr	
30	Yes	High	Fair	
31	No	Medium	Good	
32	yes	High	Fair	
<random value here>	<random>	Low	<random>	
<ra>	<ra>	Medium	<ra>	
<ra>	<ra>	high	<ra>	

originally data set has 1000 observations, now we added extra 3 rows

so total observations are $1000 + 3 = 1003$

in that one column name : Income

Income has three labels

- low: $0 + 1 = 1$
- Medium : $990 + 1 = 991$
- High: $10 + 1 = 11$

$$p(\text{low}) = \frac{1}{1003}; p(\text{Medium}) = \frac{991}{1003}, p(\text{High}) = \frac{11}{1003}$$

Before	After	deviation
$p(\text{low}) = \frac{0}{1000} = 0$	$p(\text{low}) = \frac{1}{1003} = 0.000999$	<i>Negligible</i>
$p(\text{Medium}) = \frac{990}{1000} = 0.99$	$p(\text{Medium}) = \frac{991}{1003} = 0.98$	<i>Negligible</i>
$p(\text{High}) = \frac{10}{1000} = 0.01$	$p(\text{High}) = \frac{11}{1003} = 0.0109$	<i>Negligible</i>

when we add extra observations, it gives a negligible deviation

so it avoid failing of maths

$$\frac{3}{5} * \frac{2}{5} * \frac{0.000999}{5} * \frac{2}{5} = \frac{0.01198}{625} = < \text{no zero} >$$

Upload the solution